

PAPER<i>255: PSR J0030+0451 Isolated Neutron Star — Density Regime Positive Buoyancy and F</i>neutron Dominance

Author: Daniel T. Murphy (daniel.murphy00@gmail.com)

Series: Phase 2 Session 72d — §3.x ALMA Cycle 12 Neutron Star UQFF Integrals

■# PAPER255: PSR J0030+0451 Isolated Neutron Star — Density Regime Positive Buoyancy and Fneutron Dominance

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v4.27 — Star-Magic Physics **Source:** CondensedPhysics3.py — PSRJ0030NeutronStarFUBiCalculator (Session 72d, ALMA Cycle 12) **Date:** March 2026 **Series:** Phase 2 Session 72d — §3.x ALMA Cycle 12 Neutron Star UQFF Integrals

$$F_U(r,t) = \sum_{i=1}^4 U_{\{gi\}} + U_m + U_A - U_{\{b_i\}}, \quad \text{quad } \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

$$M_J^{\text{UQFF}} = M_J^{\text{Jeans}} \cdot \text{Bigl}(1 - [SSq] \cdot \frac{B^2}{8\pi\rho c_s^2}) \cdot \text{Bigl}, \quad \text{quad } [SSq] = 0.57$$

Abstract

PSR J0030+0451 is an isolated millisecond pulsar at ~1,100 light-years, with a mass of approximately 1.4 M_{sun} confined to a radius of ~10 km ($r = 104$ m) — the compact geometry of a neutron star. This system is the **first isolated pulsar class** in the CP3 calculator, and it introduces a new UQFF regime defined by the neutron-star-density cross-section parameter $s_n \sim 10^3$, representing the degenerate nuclear density of neutron star matter.

In the ISM systems of PAPER250–254, $sn \sim 10^4$ yields $F_{\text{neutron}} = k_{\text{neutron}} \times sn = 106$ N. For PSR J0030 at $sn = 10^3$, $F_{\text{neutron}} = 10^{10} \times 10^3 = 104$ N — a difference of 53 orders of magnitude. This neutron force is the dominant UQFF term by far.

The key **uniquely rare discovery** of this paper is that despite this 53-order amplification of F_{neutron} , and despite the compact scale ($r = 104$ m vs $r = 6.17 \times 10^6$ m for the SNRs), PSR J0030 is a **positive buoyancy** system: $F_{\text{UBi}} \sim +2.53 \times 10^{28}$ N. The compact-scale geometry at $r = 10^2$ preserves the positive sign. The equivalence class extends across 14 orders of magnitude in radius and 53 orders in sn .

1. System Parameters

Parameter	Symbol	Value	Units	Source
Distance	d	~1,100	ly	Chandra/NICER 2019
Mass	M	$1.4 M_{\text{sun}} = 2.786 \times 10^{30}$	kg	Neutron star canonical
Neutron star radius	r	104	m	~10 km
X-ray luminosity	L_X	10^{31}	W	NICER 2019

Parameter	Symbol	Value	Units	Source
Surface B field	B0	108	T	Millisecond pulsar typical
System frequency	ω_0	10^{12}	rad/s	Same as SNR class
Neutron cross-section	s_n	10^3	—	NS density (vs ISM 10^4)

2. Core Physics: Neutron-Star Density Regime

2.1 F_{neutron} — The New Dominant Term

$$F_{\text{neutron}} = k_{\text{neutron}} \times s_n = 10^{10} \times 10^3 = 10^4 \text{ N}$$

For comparison:

$$F_{\text{LENR}} (\omega_0=10^{12}) \sim 6.17 \times 10^3 \text{ N}$$

$$F_{\text{neutron}} / F_{\text{LENR}} = 10^4 / 6.17 \times 10^3 \sim 1.6 \times 10^0$$

F_{neutron} exceeds F_{LENR} by 9 orders for the neutron star regime. The force hierarchy shifts from LENR-dominant (ISM and SNR systems) to **neutron-dominant** (compact objects):

$$\begin{aligned} &\text{Force hierarchy at } \omega_0=10^{12}, s_n=10^3: \\ &F_{\text{neutron}} \sim 10^4 \text{ N [dominant – 9 orders above } F_{\text{LENR}}] \\ &F_{\text{LENR}} \sim 6 \times 10^3 \text{ N [second]} \\ &F_{\text{Newt}} \sim GM/r^2 \cdot |x_2| \text{ [negligible]} \\ &F_{\text{res}} \ll F_{\text{LENR}} \text{ [DPM invisible – same conclusion as PAPER_251]} \end{aligned}$$

2.2 Compact Geometry and P Positive Buoyancy Preservation

Despite the 9-order dominance of F_{neutron} over F_{LENR} , the sign of F_{UBi} remains positive. This is because the compact geometry ($r = 104 \text{ m}$) affects the $\text{term}_{\text{gravity}} = GM/r^2$ and the integration limit x_2 in a way that preserves the positive root:

$$\begin{aligned} \text{term}_{\text{gravity}} &= G \cdot M / r^2 = 6.674 \times 10^{-11} \times 2.786 \times 10^{30} / (104)^2 \\ &\sim 1.86 \times 10^6 \text{ m/s}^2 \text{ [huge surface gravity]} \end{aligned}$$

The quadratic discriminant $b^2 - 4ac$ with $a = 1.86 \times 10^6$, $b = 4.72 \times 10^3$, $c \sim -1.83 \times 10^7$ gives a positive x_2 root (same sign as ISM systems), because the vacuum energy $F_0 = 1.83 \times 10^7 \text{ N}$ overwhelms the sign-determining coefficient c regardless of the surface gravity scale.

Key result: $x_2 > 0$? $F_{\text{UBi}} = \text{integrand} \times |x_2| > 0$? **positive buoyancy at $F_{\text{UBi}} \sim +2.53 \times 10^{20} \text{ N}$.**

2.3 53-Order s_n Range: Equivalence Class Breadth

The s_n parameter spans:

$$\begin{aligned} s_n \text{ (ISM/SNR systems): } &\sim 10^4 \text{ ? } F_{\text{neutron}} = 10^6 \text{ N [PAPER_250–254]} \\ s_n \text{ (PSR J0030): } &\sim 10^3 \text{ ? } F_{\text{neutron}} = 10^4 \text{ N [this paper]} \end{aligned}$$

53 orders of magnitude in s_n , yet both classes show **positive buoyancy at $\omega_0 = 10^{12}$** . This confirms that s_n (like B0 in PAPER_251) does not breach the equivalence class — the ω_0 parameter remains the exclusive class determinant.

2.4 DPM Resonance at $B_0 = 108 \text{ T}$

$$\begin{aligned} \text{DPM_resonance (PSR J0030)} &= 2 \cdot \mu_B \cdot B_0 / (\hbar \cdot \omega) \\ &= 2 \times 9.274\text{e-}24 \times 108 / (1.0546\text{e-}34 \times 10^{12}) \\ &\sim 1.76 \times 10^{31} \end{aligned}$$

This is an astronomically large DPM resonance, yet it is still invisible relative to F_{neutron} : $F_{\text{res}}/F_{\text{neutron}} \ll 1$. The DPM invisibility theorem (PAPER251) extends to the neutron-star-density regime.

3. Extended Force Equivalence Class Theorem

Theorem (UQFF NS-Density Class Extension): The positive buoyancy equivalence class at $\omega = 10^{12}$ rad/s includes compact objects with neutron-star densities ($s_n \sim 10^3$) in addition to diffuse ISM systems ($s_n \sim 10^4$). $F_{\text{UBi}} \sim +2 \times 10^{28}$ N regardless of s_n spanning 53 orders, confirming that s_n is not a class-determinant parameter. The vacuum energy anchor $F_0 = 1.83 \times 10^{71}$ N ensures $x_2 > 0$ for all physically observable values of s_n .

4. ALMA Cycle 12 Observational Context

PSR J0030+0451 is an ALMA Cycle 12 proposal target. Observable UQFF signatures include:

- Isotopic anomaly:** LENR neutron-capture at $F_{\text{neutron}} = 10^{47}$ N (53 orders above ISM) predicts elevated ${}^2\text{H}/{}^1\text{H} > 10^5$ and ${}^{13}\text{C}/{}^{12}\text{C} > 0.01$ in the pulsar wind nebula — detectable with ALMA Band 6 at 230 GHz.
- EHT polarimetry:** The extreme $\text{DPM_resonance} \sim 1.76 \times 10^{31}$ at $B_0 = 108$ T predicts distinctive helical B-field structure in the pulsar wind, detectable with EHT 20 μas resolution at 230 GHz.
- NICER hotspot:** PSR J0030+0451 hotspot morphology constrains the NS mass-radius relation; UQFF predicts F_{UBi} positive — consistent with a gravitationally stable bound NS (no anomalous mass loss or unbinding).

5. References

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